

Technical Document #418: Blockchain - Comprehensive Overview

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Proof of Stake (PoS) selects validators based on their stake in the network. It is more energy-efficient than Proof of Work.

Decentralized finance (DeFi) recreates traditional financial systems using blockchain technology. It enables lending, borrowing, and trading without intermediaries.

Proof of Work (PoW) is a consensus mechanism where miners solve complex puzzles to validate transactions. It is energy-intensive but secure.

Blockchain is a distributed ledger technology that records transactions across many computers. It ensures transparency and immutability.

Cross-chain technology enables communication between different blockchains. It is essential for interoperability in the blockchain ecosystem.

Non-fungible tokens (NFTs) represent unique digital assets on the blockchain. They have transformed digital ownership and art markets.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Key Takeaways:

- Tokenomics studies the economic systems of cryptocurrency tokens.
- Cross-chain technology enables communication between different blockchains.
- Consensus mechanisms ensure all nodes in a network agree on the state of the blockchain.